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State sensing of bubble jet flow based on acoustic identification of deep learning

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ABSTRACT

Sodium-cooled fast reactors (SFRs), expected to improve the efficiency of nuclear fuel utilization, use liquid sodium as a reactor coolant, which makes it possible to breed nuclear fuels while generating electricity. To increase the safety of SFRs, it is important to develop a method to detect the generation of bubble jet flow in the early stage and identify the states of it when a heat transfer tube is damaged in a steam generator. For this issue, we propose a novel state sensing method with convolutional neural networks (CNNs) and time-frequency representations (TFRs). This study consists of three phases. First, using water and air as simulant fluids to perform the proof of concept, pipe flow sound and bubble jet flow sound are acquired, each of which simulates normal and anomaly sound. Second, three TFRs are extracted from raw signals based on short-time Fourier transform (STFT), continuous wavelet transform (CWT), and synchrosqueezed wavelet transform (SWT). Third, typical CNNs including AlexNet, VGG16, and ResNet18 are introduced for the identification of pipe flow sound and three types of bubble jet flow sound. Also, t-distributed stochastic neighbor embedding (t-SNE) is applied to visualize the learning process of CNNs. As a result, the model combining ResNet18 and STFT reaches the highest accuracy of $99.20 \pm 0.31\%$ and correctly identifies 1984 out of 2000 test data. These results demonstrate that our proposed method based on the acoustic identification of deep learning has great potential to sense the states of bubble jet flow in actual SFRs.

KEY WORDS: Two-phase flow, Bubble jet flow, Acoustic identification, Deep learning

1. INTRODUCTION

Nuclear power generation attracts high expectations toward the realization of carbon neutrality. Sodium-cooled fast reactors (SFRs), one of the next-generation reactors, are expected to improve the efficiency of nuclear fuel utilization because of their property to breed nuclear fuels while generating electricity [1]. To operate SFRs safely, it is important to detect the generation of bubble jet flow with the sodium-water reaction in the early stage and identify the states of it when a heat transfer tube is damaged in a steam generator (SG). Accidents with the sodium-water reaction in SFRs were reported in past, and dozens of heat transfer tubes were damaged in some cases [2, 3]. However, serious accidents can be prevented by early-stage detection of bubble jet flow, and some studies focus on acoustic methods with the consideration of their advantage of a short response time [4].

Recently, deep learning has been developing at an extraordinary speed with the breakthrough of computers. Convolutional neural networks (CNNs), a kind of deep learning, have the characteristics of robustness and invariance to transition, scaling, tilting, or other forms of image distortion [5]. Thus, CNNs are widely used in the field of image recognition, and many studies perform acoustic identification based on CNNs. For example, Zhang et al. performed the acoustic identification of bubble flow regimes based on CNNs and time-frequency representations (TFRs) extracted by short-time Fourier transform, continuous wavelet transform (CWT), and synchrosqueezed wavelet transform (SWT) [6]. Also, Ma et al. developed a bearing

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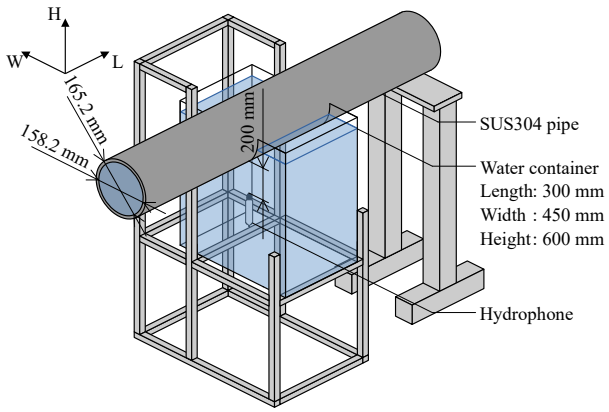


Fig. 1 Experimental arrangement for measurement of pipe flow sound.

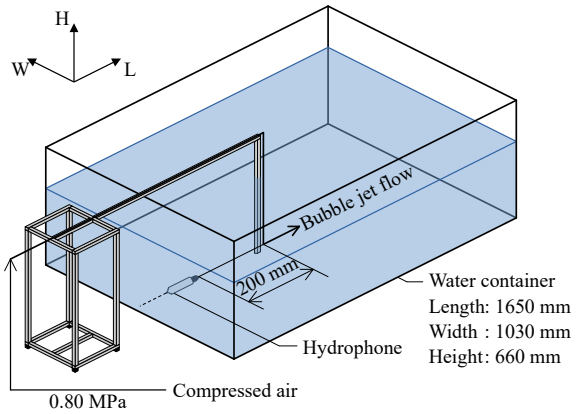


Fig. 2 Experimental arrangement for measurement of bubble jet flow sound.

fault diagnosis method based on TFRs and transfer learning-CNNs, and they visualized the feature learning process of CNNs by t-distributed stochastic neighbor embedding (t-SNE) [7]. In terms of the acoustic identification of bubble jet flow with deep learning, we mainly proposed a method using brief deep learning models and basic features such as 1D values of the spectrum and the root mean square in the previous study [8]. However, there are few examples to apply CNNs and TFRs including STFT, CWT, and SWT to the acoustic identification of bubble jet flow and visualize the feature learning process of CNNs.

Based on the above, this study covers the accidental generation of bubble jet flow in the accident of SG tubes damaging in SFRs. The main objectives of this study are to develop a novel state sensing method of bubble jet flow based on CNNs and TFRs and to visualize the feature learning process of CNNs. Our proposed method has the potential for the detection of anomalies in the early stage by the acoustic method and the detailed analysis of the accidents by deep learning, which can lead to the safe operation of SFRs. This paper begins with the description of experiments on the acquisition of pipe flow sound and bubble jet flow sound, each of which simulates the normal and anomaly sound. This is followed by the extraction of STFT, CWT, and SWT from raw signals. Then, typical CNNs including AlexNet, VGG16, and ResNet18 are applied to the acoustic identification of bubble jet flow. Finally, we discuss the accuracy of each method combining CNNs and TFRs, and we visualize the feature learning process of CNNs using t-SNE.

2. EXPERIMENTS ON SOUND ACQUISITION

Figure 1 shows the experimental arrangement for the measurement of pipe flow sound, which simulates the normal sound from SGs. The structural-acoustic coupling such as observed in this experiment [9] is assumed to be dominant in actual SFRs. As shown in Figure 1, a pipe made of SUS304 is supported horizontally and filled with water. A hydrophone (ACO TYPE7800, Japan) is located 200 mm below the lower end of the pipe in a water tank. This hydrophone has the flat frequency response and omni-directional, which enables the recording with constant sensitivity regardless of the hydrophone position or the frequency of sound signals. Also, the axial distance from the pipe inlet to the hydrophone is about 15 times that of the inner diameter of the pipe. The hydrophone records pipe flow sound at a sampling rate of 44.1 kHz for 1 minute. For sound acquisition under conditions close to the Reynolds number of actual SGs in SFRs ($Re = 2.7 \times 10^5$), the water flow rate inside the pipe is set to be 125 m^3/h ($Re = 3.2 \times 10^5$).

Figure 2 shows the experimental arrangement for the measurement of bubble jet flow sound, which simulates the anomaly sound from SGs. Air compressed to 0.80 MPa (absolute pressure) is released from a nozzle into water. Three types of nozzles are prepared, each with a nominal diameter of 0.15 mm, 0.17 mm, and 0.20 mm. The hydrophone same as mentioned previously is located 200 mm behind of bubble jet flow. It

records bubble jet flow sound at a sampling rate of 44.1 kHz for 1 minute in each case of three nozzles. In all cases, the flow is choked at the outlet of nozzles, consistent with the condition in the actual SFR accident.

3. EXTRACTION OF 2D TIME-FREQUENCY REPRESENTATIONS

3.1 Theory of Time-Frequency Analysis

In the present study, features input to CNNs are TFRs, which have three components: time, frequency, and signal intensity. TFRs are in the form of heat maps, and the signal intensity is represented in color. In this section, raw signals are converted to TFRs based on three methods of time-frequency analysis including STFT, CWT, and SWT. Each theory is described below.

First, given a time signal $s(t)$, the STFT is defined as [10]:

$$\mathrm{STFT}(t,\omega) = \int_{-\infty}^{\infty} s(\tau)h(\tau - t)e^{-j\omega\tau}d\tau \tag{1}$$

where $h(t)$ is a window function, and ω denotes an angular frequency. Also, we introduce the hanning window as a window function. As shown in Eq. 1, the STFT calculates the Fourier transform of products of the time signal and time-shifted window function. The STFT is suitable for periodic signals because its basic function is trigonometric. Second, the CWT is defined as follows:

$$\mathrm{CWT}(a,b) = \frac{1}{\sqrt{a}}\int_{-\infty}^{\infty} s(t)\bar{\psi}\left(\frac{t-b}{a}\right)dt \tag{2}$$

where a and b denote the scale and time-shift parameters, respectively. Also, $\bar{\psi}(t)$ is the complex conjugate of a mother wavelet $\psi(t)$, and the complex Morlet wavelet is applied to the mother wavelet. The CWT is a method to represent original signals by the superposition of mother wavelets. Differently from the STFT, the CWT is also effective to process signals with aperiodicity or drastic changes of frequency components because its basic function is aperiodic. Assume that a signal is purely harmonic as $s(t) = A\cos(\omega t)$, the CWT can be expressed as follows using Plancherel's theorem:

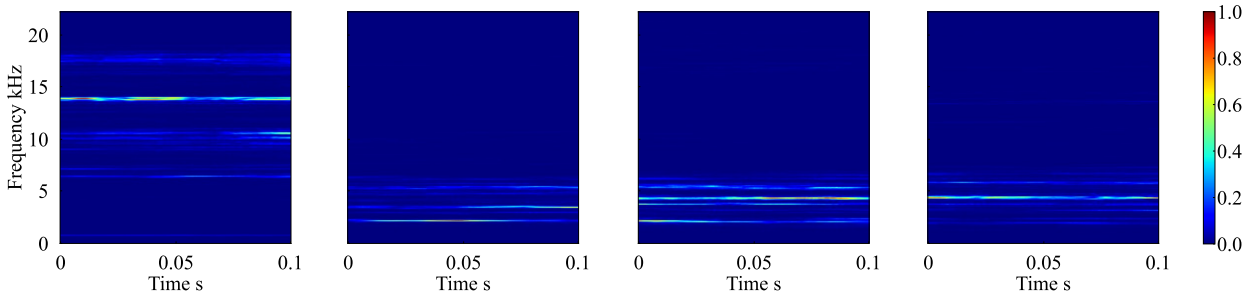
$$\begin{aligned}\mathrm{CWT}(a,b) &= \frac{\sqrt{a}}{2\pi}\int_{-\infty}^{\infty}\hat{s}(\varepsilon)\bar{\hat{\psi}}(a\varepsilon)e^{jb\varepsilon}d\varepsilon \\ &= \frac{A\sqrt{a}}{4\pi}\int_{0}^{\infty}[\delta(\varepsilon-\omega)+\delta(\varepsilon+\omega)]\bar{\hat{\psi}}(a\omega)e^{jb\varepsilon}d\varepsilon \\ &= \frac{A\sqrt{a}}{4\pi}\bar{\hat{\psi}}(a\omega)e^{jb\omega}\end{aligned} \tag{3}$$

where $\hat{s}(\varepsilon)$ and $\bar{\hat{\psi}}(\varepsilon)$ denote the Fourier transform of a time signal $s(t)$ and the complex conjugate of the Fourier transform of a mother wavelet $\psi(t)$, respectively. Also, δ is the delta function. Equation 3 demonstrates that $\mathrm{CWT}(a,b)$ is spread out over a region around $a = \omega_0/\omega$ if $\hat{\psi}(\varepsilon)$ is concentrated around $\varepsilon = \omega_0$; the signal intensity is blurred in the scale direction. The SWT is a method to improve this issue and it is defined as shown in the following equation [11]:

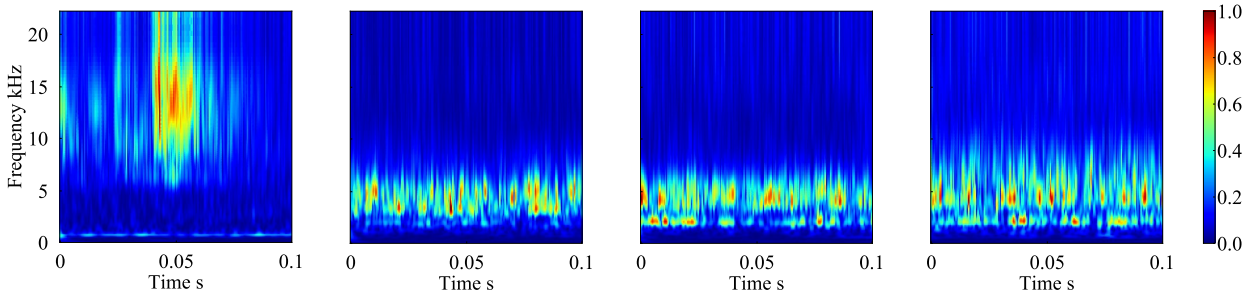
$$\mathrm{SWT}(\omega_l,b) = \frac{1}{\Delta\omega}\sum_{a_k:|\omega(a_k,b)-\omega_l|\le\Delta\omega/2}\mathrm{CWT}(a_k,b){a_k}^{-3/2}(\Delta a)_k \tag{4}$$

where $\omega(a,b)$ is a candidate instantaneous angular frequency calculated as:

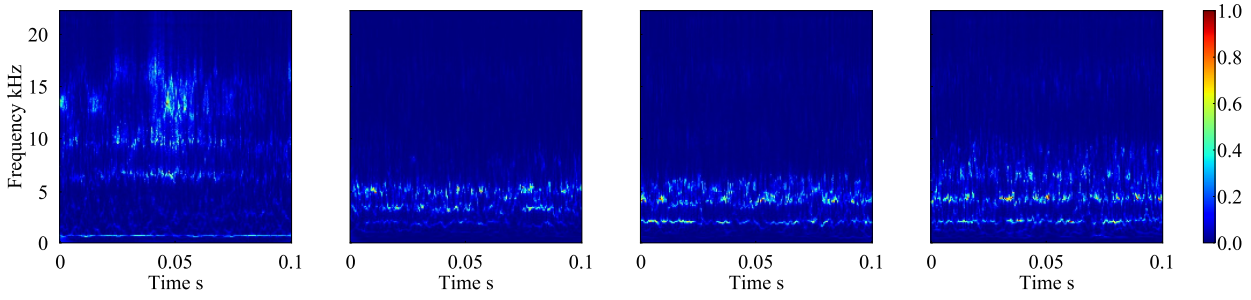
$$\omega(a,b) = -j\frac{1}{\mathrm{CWT}(a,b)}\frac{\partial}{\partial b}\mathrm{CWT}(a,b) \tag{5}$$



(a) STFT: short-time Fourier transform



(b) CWT: continuous wavelet transform



(c) SWT: synchrosqueezed wavelet transform

Fig. 3 Examples of extracted TFRs: (a)STFT, (b) CWT, and (c) SWT (from left to right: pipe flow sound, bubble jet flow sound from nozzles with diameters of 0.15, 0.17, and 0.20 mm).

and $(\Delta a)_k = a_k - a_{k-1}$, $\Delta\omega = \omega_l - \omega_{l-1}$. As shown in Eq. 4, the signal intensity is calculated by squeezing values in nearby interval $[\omega_l - \Delta\omega/2, \omega_l + \Delta\omega/2]$ to $\omega(a_k, b)$ at a_k. In the SWT, the blurring region in the CWT is squeezed into a concentrated line, which improves the time-frequency resolution [6].

3.2 Extracted 2D Time-Frequency Representations

Examples of extracted TFRs are shown in Fig. 3. From raw signals, TFRs are extracted to have time information for 0.1 s and frequency information between 0 Hz to 44.1/2 kHz. Focusing on the distribution of signal intensity, the pipe flow sound has several peaks in the frequency band above 5000 Hz. These peaks can be described by coupled waves of cylinder walls and internal liquid. On the other hand, three types of bubble jet flow sound have a characteristic frequency band of about 1500 Hz to 5000 Hz. We speculate that the bubble jet flow sound is caused by the vibration, coalescence and collapse of bubbles, and the release of bubbles from the nozzle. Also, we can see the difference in the signal intensity distribution is not obvious between the three types of bubble jet flow sound, which indicates that these sounds are difficult to be classified. In addition, comparing the CWT and SWT, the signal intensity is blurred in the scale direction for the CWT, whereas it is concentrated in the instantaneous frequency for the SWT. This result agrees

Table 1 Dataset of three TFRs prepared for deep learning.

	Pipe flow	Bubble jet flow 0.15 mm	Bubble jet flow 0.17 mm	Bubble jet flow 0.20 mm
Size of training dataset	450	450	450	450
Size of validation dataset	100	100	100	100
Size of test dataset	50	50	50	50

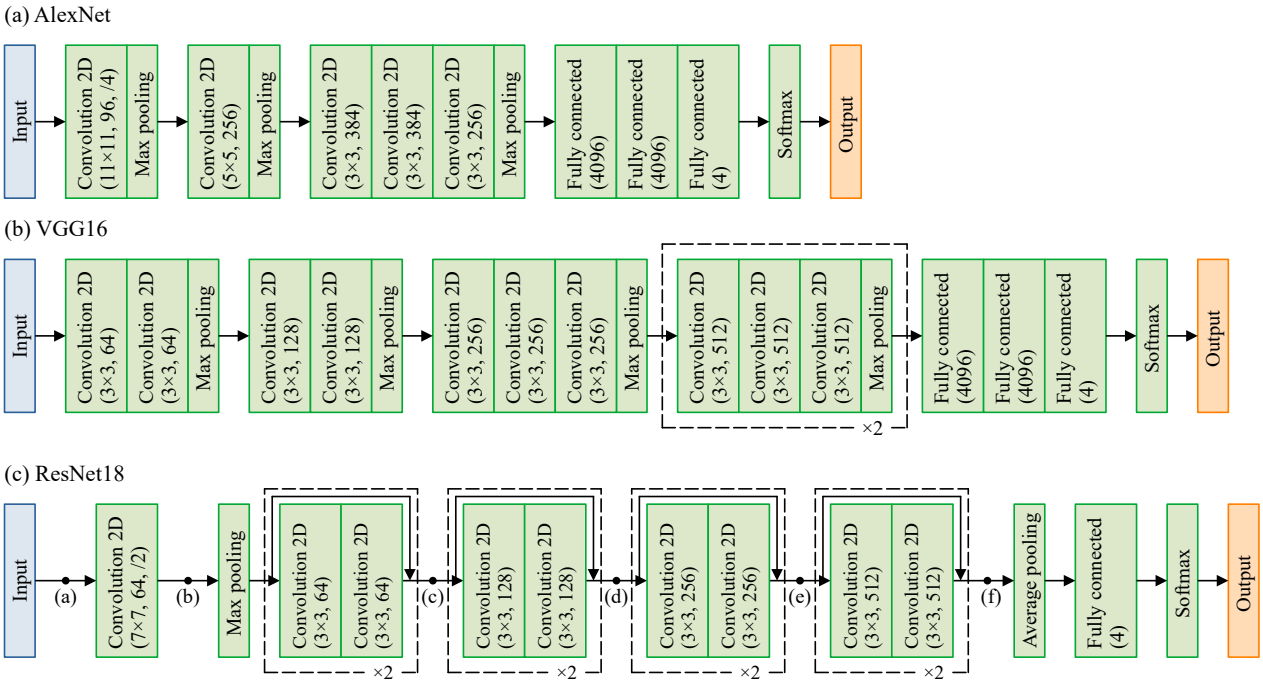


Fig. 4 Schematic diagrams of typical CNNs: (a) AlexNet, (b) VGG16, and (c) ResNet18.

with the theory mentioned in Section 3.1, and we can confirm the improvement in frequency resolution by squeezing.

We generate the dataset for deep learning as shown in Table 1. Four types of sound are prepared, and each has an amount of 600 data. It is divided into 450 data for the training dataset, 100 data for the validation dataset, and 50 data for the test dataset. Also, the shape of 2D images is set to $224 \times 224 \times 3$, which means that each 2D image has 224 values in width, 224 values in height, and 3 values in RGB scale.

4. CNN-BASED ACOUSTIC IDENTIFICATION OF BUBBLE JET FLOW

4.1 CNN Models

For the acoustic identification of bubble jet flow, AlexNet [12], VGG16 [13], and ResNet18 [14] are introduced to evaluate the feasibility of our proposed method using their typical ability. Figure 4 shows the schematic diagrams of them. First, the AlexNet, the winner of the ImageNet Large Scale Visual Recognition (ILSVRC) 2012, consists of five convolution layers and three fully connected layers. It made a breakthrough in the development of CNN architectures for image recognition, and its methods including the rectified linear unit (ReLU) function and the dropout became standard in the construction of CNN architectures. Second, the VGG16 took second place in ILSVRC 2014. It mainly consists of thirteen convolution layers and three

Table 2 Adjusted hyperparameters for learning of three typical CNNs.

CNN model	Batch size	Epoch	Learning rate	Optimizer	Loss function
AlexNet	5	100	1.0×10^{-6}	Adam	Cross entropy loss function
VGG16			5.0×10^{-7}		
ResNet18			1.0×10^{-5}		

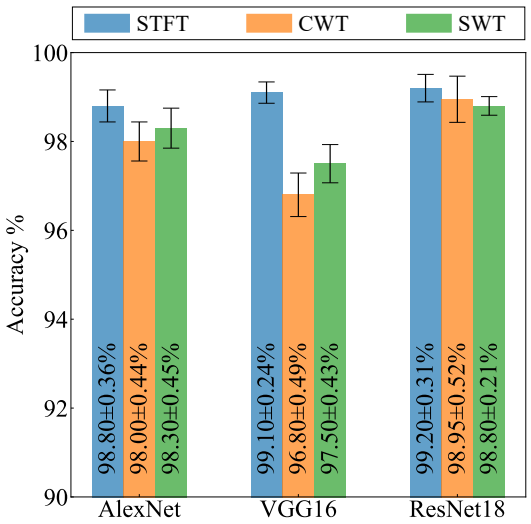


Fig. 5 Identification accuracy of each method combining three CNNs and three TFRs.

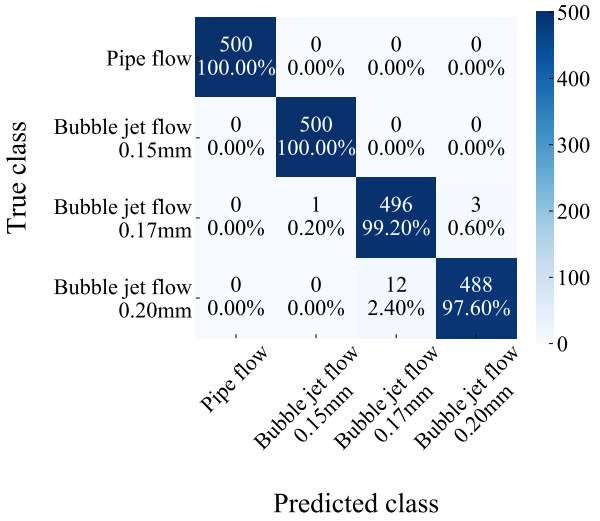


Fig. 6 Confusion Matrix of the classification result of the method combining ResNet18 and STFT.

fully connected layers. Its characteristic is the repetition of the convolution with small kernels and the max-pooling. Although the deepening of layers improves accuracy, the learning of CNNs sometimes does not work well because of the difficulty in propagating errors to the upper layers. The ResNet, the winner of ILSVRC 2015, overcomes this problem through the introduction of a shortcut connection. The process including the shortcut connection is called the residual block, which outputs the sum of an input x and the transformation $f(x)$. This makes it easier to propagate errors to the upper layers and enables very deep CNNs to learn efficiently. We introduce the ResNet18, which includes eight residual blocks and consists of seventeen convolution layers and one fully connected layer. In the present study, the initial parameters of the CNNs are randomly given. Also, the CNNs are constructed to classify four types of sound: pipe flow sound, and bubble jet flow sound from leakage holes with diameters of 0.15 mm, 0.17 mm, and 0.20 mm.

4.2 Results and discussions

The CNNs are constructed, trained, validated, and tested with the TensorFlow framework. The adjusted hyperparameters including the batch size, the number of epochs, and the learning rate are shown in Table 2. These parameters are determined based on the history of the accuracy and loss in the validation phase. Also, this study utilizes the Adam as an optimizer and the cross-entropy loss function as a loss function. In addition, we perform the K-fold cross validation [15], in which the total data is divided into K groups, and each group is used as the test data once. This method makes it possible to evaluate the generalization performance of CNNs by reducing the influence of the test data bias. In the present study, we set the value of K to 10; the ratio of the training data to the test data is $9:1$.

Figure 5 shows the identification accuracy of each method combining three CNNs and three TFRs. Every method achieves more than 96.80%, and the identification accuracy does not change significantly between

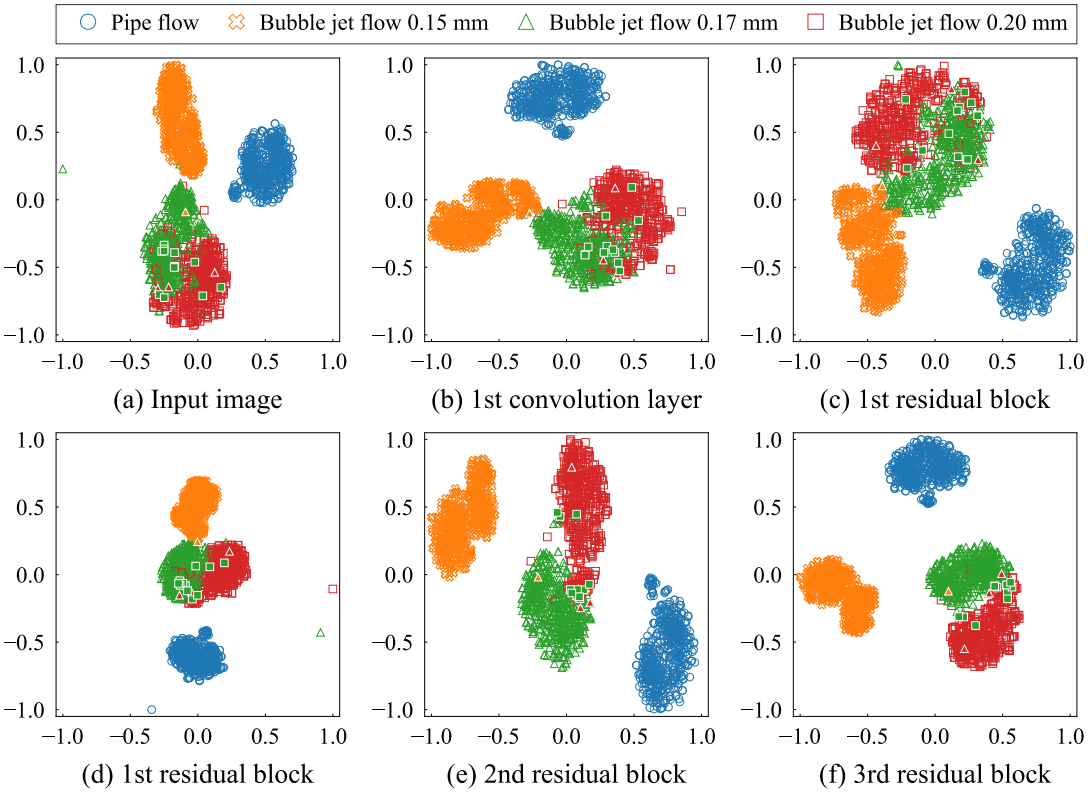


Fig. 7 Feature learning process of the method combining ResNet18 and STFT visualized via t-SNE (True and predicted classes are represented by shape and color, respectively. Additionally, misclassified data are filled in colors of prediction classes for clarification).

several methods. This demonstrates that our proposed method based on typical CNNs and TFRs is suitable for the identification of the pipe flow and the bubble jet flow. Also, the method combining ResNet18 and STFT reaches the highest accuracy of 99.20% and the standard error is only 0.31%, which shows this method can identify four types of sound with high accuracy and stability. The classification result of the method combining ResNet18 and STFT is shown as a confusion matrix in Fig. 6. In the confusion matrix, horizontal and vertical directions represent the true and predicted classes, respectively. First, this method can distinguish between pipe flow sound and bubble jet flow sound with an accuracy of 100.00%, which demonstrates that this method has the possibility to detect the accidental generation of bubble jet flow in actual SFRs. In addition, although this method makes faults in the classification of three types of bubble jet flow, 1984 out of 2000 data are identified as correct classes. This result shows that this method has the potential to analyze the progression of accidents with high accuracy in actual SFRs; the sodium-water reaction widens the leakage hole in actual SFR accidents.

The algorithm for dimensionality reduction t-SNE [16] is introduced to visualize the learning process of CNNs. In the process, the similarity p_{ij} of high dimensional datapoints x_i and x_j is given as follows:

$$p_{ij} = \frac{p_{i|j} + p_{j|i}}{2n}, \quad p_{j|i} = \frac{\exp(-||x_i - x_j||^2/2{\sigma_i}^2)}{\sum_{k \neq i} \exp(-||x_i - x_k||^2)/2{\sigma_i}^2} \tag{6}$$

where n is the amount of high dimensional datapoints, and σ_i is the variance of the Gaussian that is centered on the datapoint x_i. Based on the Student t-distribution, the similarity q_{ij} of the low dimensional datapoints

y_i and y_j is given as follows:

$$q_{ij} = \frac{(1 + ||y_i - y_j||^2)^{-1}}{\sum_{k \neq i}(1 + ||y_i - y_j||^2)^{-1}} \tag{7}$$

The difference between the distance of high dimensional datapoints and that of low dimensional datapoints is described by a single Kullback-Leibler divergence as a loss function:

$$C = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}} \tag{8}$$

To minimize C, the low dimensional datapoints are updated by the stochastic gradient descent (SGD), using the gradient of C:

$$\frac{\delta C}{\delta y_i} = 4 \sum_j (p_{ij} - q_{ij})(y_i - y_j)(1 + ||y_i - y_j||^2)^{-1} \tag{9}$$

Through this process, the high dimensional data is transformed into the low dimensional data, usually 2D or 3D. Figure 7 shows the feature learning process of the method combining ResNet18 and STFT visualized by t-SNE. In each visualization result, the horizontal and vertical axes denote the first and second components of the low dimensional datapoints. In the present study, t-SNE is utilized into the input images, the outputs of the first convolution layer, and the last outputs of each residual block (The alphabetical numbers (a) to (f) in Fig. 7 correspond to those in Fig. 4). First, the visualization of input images shows that the points in the same class are mostly clustered at this phase although there is an overlapping region between the bubble jet flow sound from the leakage holes of 0.17 mm and 0.20 mm, which demonstrates the effectiveness of STFT as a feature extraction method. Second, as the layers go deeper, the points of different classes are more separable, which indicates that ResNet18 learns the features of STFT adaptively. In addition, looking at the visualization of the outputs of the 4th residual block, the clusters of the pipe flow sound and the bubble jet flow sound from the leakage hole of 0.15 mm are separated completely, and the overlapping region between the bubble jet flow sound from the leakage holes of 0.17 mm and 0.20 mm decreases significantly. This result agrees with the confusion matrix of the classification result as shown in Fig. 6; this method makes faults in the classification of the bubble jet flow sound from the leakage holes of 0.17 mm and 0.20 mm.

We have developed a novel state sensing method of bubble jet flow based on CNNs and TFRs. This method has the potential to detect the generation of bubble jet flow in the early stage and to analyze its progression with high accuracy, which leads to the safe operation of SFRs. Although the STFT achieves the highest accuracy in the present study, the CWT and SWT may surpass it under unsteady and noisy environments. Also, the passive acoustic method employed in this study can not work well depending on the locations of sensors in the situation where sound propagation is impeded by the dominant bubbles. Since this study does not cover this situation, we also have to develop an active acoustic method as a complementary measure that evaluates the attenuation of sound intensity by passing through bubbles.

5. CONCLUSION

The present study has covered the accidental generation of bubble jet flow caused by the SG tube damaging in SFRs. We have developed a novel state sensing method of bubble jet flow based on typical CNNs and TFRs and visualized the learning process of it using t-SNE. First, pipe flow sound and bubble jet flow sound have been measured using air and water as simulant fluids to perform the proof of concept, each of which simulates the normal and anomaly sound from the SG. Second, three types of TFRs generated based on STFT, CWT, and SWT have been extracted from raw signals. Third, three typical CNNs including AlexNet, VGG16, and ResNet18 have been constructed to classify the four types of sound. Through the discussion of CNNs performance and the visualization of the learning process, our findings are summarized as follows:

- Every method combining CNNs and TFRs classifies the four types of sound with an accuracy of more than 96.80%, which indicates that our proposed method based on acoustic identification and deep learning is suitable for the state sensing of bubble jet flow.
- The method combining ResNet18 and STFT reaches the highest accuracy of $99.20 \pm 0.31\%$ and identifies 1984 out of 2000 data as the correct classes. The results show that this method has the possibility to detect the generation of bubble jet flow and to analyze the progression of it with high accuracy in actual SFR accidents.
- Through visualization of the learning process via t-SNE, ResNet18 learns the features of STFT adaptively as the layers go deeper. Also, the visualization result of the outputs from the fourth residual block agrees with the classification result of the confusion matrix.

In future works, the accuracy generally becomes higher as the dataset size increases [6], therefore we plan to improve the identification accuracy of CNNs through the increase of the dataset size. Also, although this study is in the phase of the proof of concept, it is necessary to perform experiments under conditions close to those in actual SFRs such as the sound acquisition of the sodium-water reaction and the two-phase jet flow under higher pressure; the two-phase jet flow involves the sodium-water reaction and it is released at a pressure of several tens atmospheres in actual SFRs.

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NOMENCLATURE

a	scale parameter	q_{ij}	similarity of low dimensional datapoints
A	amplitude of purely harmonic signal	s	signal
b	time-shift parameter	t	time
C	single Kullback-Leibler divergence	x_i, x_j	high dimensional datapoints
h	window function	y_i, y_j	low dimensional datapoints
j	imaginary unit	δ	delta function
K	number of K-fold cross validation	σ_i	variance of the Gaussian
n	number of high dimensional datapoints	ψ	mother wavelet
p_{ij}	similarity of high dimensional datapoints	ω	angular frequency

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